

Ch 8 Techniques of Integration

The Fundamental Theorem tells us how to evaluate a definite integral once we have an antiderivative for the integrand function. However, finding antiderivatives (or indefinite integrals) is not as straightforward as finding derivatives. In this chapter we study a number of important techniques that apply to finding integrals for specialized classes of functions such as trigonometric functions, products of certain functions, and rational functions. Since we cannot always find an antiderivative, we develop numerical methods for calculating definite integrals. We also study integrals whose domain or range are infinite, called *improper integrals*.

8.1 Using Basic Integration Formulas

① Basic Integration Formulas

- (a) $\int k \, du = ku + C$
- (b) $\int u^n \, du = \frac{1}{n+1} u^{n+1} + C, \, n \neq -1$
- (c) $\int \frac{1}{u} \, du = \ln|u| + C$
- (d) $\int e^u \, du = e^u + C$
- (e) $\int \sin u \, du = -\cos u + C, \quad \int \cos u \, du = \sin u + C$
- (f) $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\left(\frac{u}{a}\right) + C, \quad (u^2 < a^2)$
- (g) $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

② Rewriting an Integral to Match it to a Standard Formula

(a) Making a substitution

Example 1 Evaluate $\int \frac{2x-9}{\sqrt{x^2-9x+1}} \, dx$.

Example 2 Evaluate $\int \frac{1}{\sqrt{1-4x^2}} dx$.

Example 3 Evaluate $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$.

Example 4 Evaluate $\int \frac{4}{1+4x^2} dx$.

Example 5 Evaluate $\int \frac{1}{e^x + e^{-x}} dx$.

Example 6 Evaluate $\int \frac{1}{\sqrt{9-x^2}} dx$.

(b) Completing the square

Example 7 Evaluate $\int \frac{1}{\sqrt{8x-x^2}} dx$.

Example 8 Evaluate $\int \frac{1}{x^2+2x+2} dx$.

(c) Reducing an improper function

Note: The integrand is an improper fraction if the degree of the numerator is greater than the degree of the denominator.

Example 9 Evaluate $\int \frac{2x}{x+1} dx$.

Example 10 Evaluate $\int \frac{x^2+2x}{x+1} dx$.

Example 11 Evaluate $\int \frac{x^2 + 3x + 5}{x + 1} dx$.

Example 12 Evaluate $\int \frac{3x + 2}{\sqrt{1 - x^2}} dx$.

Example 13 Evaluate $\int \frac{3x^2 - 7x}{3x + 2} dx$.

Example 14 Evaluate $\int \frac{4x^3 - x^2 + 16x}{x^2 + 4} dx$.

Example 15 Evaluate $\int \frac{2x^4 + x^2 + 3}{x^2 + 1} dx$.